

OMP FPGA Project — HDL-B Status & 3-Day Team Plan

Three working days remain: Fri / Sat / Sun.

Part 1 — What HDL-B finished (the simulator)

The bit-accurate Python fixed-point reference is **done and verified**. This is the critical-path deliverable: it is the thing the contract (§9) names as the source of `expected_idx` / `expected_val`, and it is what HDL-A's RTL gets validated against.

What was built and confirmed:

- `src/omp_params.py` — canonical Python copy of the frozen §2 constants (WINDOW_N=256, M_LEN=128, N_ATOMS=256, Q1.15 I/O, 100 MHz). Single source of truth; no magic numbers elsewhere.
- `src/omp_sim.py` — `omp_argmax_fixed(D_q15, r_q15, mac_bits)`. Models one hardware COMPUTE pass (§8): integer inner product of every atom against the residual in an int64 accumulator (Q8.30 behavior per §3), argmax of the absolute value. No floating point after input promotion → bit-exact.
- `tests/test_golden.py` — pytest regression suite. Parametrized over all five golden vectors; asserts both `idx` and `val` match bit-for-bit. Fails loudly (not skips) if the vectors can't be found.

Validation result — **all 5 vectors pass on both index and accumulator value**:

Vector	idx	val	margin	what it tests
test_01_easy_margin	PASS	PASS	78.0%	clear winner (§9a)
test_02_tight_margin	PASS	PASS	0.0023%	bit-accuracy under a near-tie (§9b)
test_03_winner_near_index_0	PASS	PASS	72.0%	no off-by-one at low index (§9c)
test_04_winner_near_index_N	PASS	PASS	77.4%	no off-by-one at high index (§9d)
test_05_engineered_tie	PASS	PASS	exact tie	tie-break convention (§9e)

The tight case (winner beats runner-up by ~0.002%) is the strongest evidence the arithmetic is correct — any bit-level error would flip the winner. The tie case (atoms 10 and 200 exactly equal) is confirmed against the agreed rule.

Two interface details locked with HDL-A (record per §10):

- **MAC_BITS truncation:** truncate-before-multiply. *Confirmed by HDL-A.*
- **Argmax tie-break:** lowest index wins. *Confirmed by HDL-A.*

Immediate housekeeping (do before touching anything else)

1. **Push to the shared repo.** All of the above currently lives only on the local machine. Until pushed, HDL-A cannot run the test against their RTL and the work does not count for the team.
2. **Post the §9-closing message:** "HDL-B sim validated bit-exact against all 5 golden vectors in [goldenVectors/](#). Tie-break = lowest index, truncation = before-multiply (confirmed w/ HDL-A). Regression test in [HDL-B/tests/](#). Pushed."
3. **Agree the vectors folder name** ([goldenVectors/](#) at repo root) so HDL-A's testbench and HDL-B's test point at the same path. This was a divergence risk.

Part 2 — The Next Three Days

A *fully integrated* system — synthesizable RTL through Vivado place-and-route, an AXI driver on the Cortex-A9, cycle-accurate timing measured on the Zybo board, and a tuned software baseline to compare against — is **unlikely to be finished and debugged in three days from a standing start**. Hardware bring-up (AXI address mapping, BRAM loading, timing closure, "why does STATUS never read DONE") routinely eats more time than the logic itself.

So do not plan the poster around the board. Plan it around what you can produce with high confidence, and treat the live hardware as a stretch goal.

The key insight: the poster's headline does not require a working board

The proposal's headline contribution is the **bit-width Pareto front**: reconstruction quality vs. resource cost across MAC_BITS. Its three axes come from three places:

Axis	Source	Needs working hardware?
PRD (accuracy)	HDL-B simulator sweep	No — pure simulation
Resources (LUT/DSP/BRAM)	HDL-A <i>synthesis</i> reports	No — synthesis estimates only

Latency (cycles)

HDL-A FSM cycle count
(analytical)

No — countable from the
design

Every axis of your headline plot can be produced from **simulation + synthesis reports**, with no signal ever running on the physical Zynq. The Cortex-A9 baseline and a live demo are *nice-to-haves*, not the headline. Protect the headline; let the rest be stretch.

Part 3 — Poster MVP (what MUST land by Sunday)

Define the floor first, then add if time allows.

Tier 1 — Must have (the poster is viable with only this):

- The OMP algorithm + system architecture diagram (already drafted in the proposal).
- PRD vs. MAC_BITS curve from the HDL-B simulator on real MIT-BIH data.
- Resource (DSP / LUT / BRAM) vs. MAC_BITS from HDL-A synthesis reports.
- The combined **Pareto front**: quality vs. resource cost, the headline figure.
- A statement of which OMP stage tolerates low precision and which doesn't.

Tier 2 — Strong if achieved:

- Analytical latency (cycles → μs at 100 MHz) per operating point.
- RTL validated against the 5 golden vectors (proves the design is *correct*, even if not yet running on the board).

Tier 3 — Stretch (only if hardware integration genuinely lands):

- Measured on-board latency vs. the analytical estimate.
- Cortex-A9 software baseline and the hardware-vs-software speedup number.
- Live demo / photo of the board reconstructing an ECG beat.

If Tier 3 slips, the poster is still complete and tells a coherent story. Make sure everyone understands that *before* Friday, so nobody burns Sunday night chasing a demo at the expense of the Pareto plot.

Part 4 — Task breakdown by person

HDL-B (you) — you own the entire accuracy axis; you can fully deliver it solo

1. **Push the sim + close §9** (above). ~30 min.
2. **Phase 4 — OMP loop wrapper.** Wrap `omp_argmax_fixed` in a full OMP loop: the fixed-point argmax is the inner step; the least-squares solve + residual update run in float/high precision (that part runs on the Cortex-A9 in the real system, so it is *not* the bit-width variable and need not be quantized). Output: a reconstructed signal from MIT-BIH windows.
3. **Phase 5 — the sweep.** Run the loop over MIT-BIH windows for $\text{MAC_BITS} \in \{8, 10, 12, 14, 16\}$; record PRD per setting. Produce the PRD-vs-MAC_BITS curve. **This is the entire accuracy axis of the headline plot and needs nobody else.**
4. Hand HDL-A the exact MAC_BITS list so synthesis runs match your sweep points.

HDL-A — owns the resource + latency axes

1. **Synthesizable correlation-engine RTL** (the §8 COMPUTE argmax — that is the whole hardware job; LS is software). Validate against the 5 golden vectors using HDL-B's expected values.
2. **Synthesize at each MAC_BITS sweep point** and pull LUT / DSP / BRAM from the utilization report. This alone produces the resource axis — *no board required*.
3. Report the FSM cycle count → analytical latency. `~N_ATOMS + pipeline_depth`.
4. *Stretch:* place-and-route + on-board bring-up with SW.

SW — owns the baseline + integration glue

1. **AXI driver** per §4/§6 (BRAM load, START, poll DONE, read RESULT_IDX). Needed for any on-board run.
2. **Cortex-A9 float OMP baseline** with cycle-accurate timing — the speedup comparison. This is independent of the FPGA and can be written and measured *today* on the PS.
3. *Stretch:* integrate driver + HDL-A's RTL on the board for end-to-end numbers.

Part 5 — Day-by-day

	HDL-B	HDL-A	SW
Fri	Push sim, close §9. Build + sanity-check Phase 4 OMP loop on one MIT-BIH record.	Write correlation-engine RTL; validate vs. golden vectors.	Write Cortex-A9 baseline OMP; get it compiling and timing on the PS.
Sat	Run full MAC_BITS sweep → PRD curve. Hand sweep points to HDL-A.	Synthesize at each sweep point; collect LUT/DSP/BRAM. Report cycle count.	Finish baseline timing. Start AXI driver. (Stretch: begin board bring-up.)

Sun Assemble Pareto front from your PRD + HDL-A's resources. Write the precision-tolerance findings. Build poster figures.	Hand resource table to HDL-B. (Stretch: P&R + board.)	(Stretch: integration.) Otherwise: baseline number into the poster.
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The Friday hand-offs matter: HDL-B's golden-vector validation is what lets HDL-A trust their RTL, and HDL-B's sweep-point list is what makes HDL-A's synthesis runs line up with the PRD curve. Coordinate the MAC_BITS list explicitly so the two axes share an x-axis.

Part 6 — The poster data table (who fills which cell)

This is the raw table behind the Pareto plot. Fill it as numbers arrive.

MAC_BITS	PRD (%) [HDL-B]	DSP [HDL-A]	LUT [HDL-A]	BRAM [HDL-A]	Latency cyc [HDL-A]
3	82.9	0	867	16	1033
4	74.6	0	1473	16	1033
5	39.3	0	1760	16	1033
6	22.5	0	2078	16	1033
8	20.4	0	3034	16	1033
10	20.1	0	4503	16	1033
11	20.1	47	610	16	1033
12	20.1	47	610	16	1033
16	20.1	47	610	16	1033

Software baseline (separate, SW): Cortex-A9 float OMP — reconstruction time per window, PRD. The headline speedup = (baseline time) / (hardware latency at chosen MAC_BITS).

Part 7 — One risk to name out loud now

Decide *as a team, Friday morning*, that:

- The Pareto front (sim PRD + synthesis resources) is the committed deliverable.
- The board demo and SW baseline are stretch.
- The MAC_BITS sweep list is agreed and shared so the axes align.

If you align on that one paragraph, the poster lands even in the worst realistic case.